

06

Electric Vehicle
Codec Technologies India
Internship 2026

IoT EV Health Monitoring System

NodeMCU / Raspberry Pi node,
ThingSpeak or Firebase cloud,
responsive web dashboard - run
here on a simulator and mock
cloud

AT A GLANCE

370,332 B flash, 35 %

Firmware compile (recorded)

960 / 960

Simulated uploads / read back

43.76 - 51.36 V

Pack voltage range (sim)

-10.1 ... +37.5 A

Current range (sim)

0.23 %

SOC estimate mean abs error

44 of 960 samples

Alerts raised

LIVE

Dashboard status (headless)

17 / 17

Tests re-run here

VERDICT

TESTS 17/17

RE-RUN HERE

ALL DATA SIMULATED

HARDWARE NOT RUN

FIRMWARE COMPILED

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1. Objective and brief mapping

Brief: build a dashboard showing real-time EV health data using a NodeMCU or Raspberry Pi with voltage, current and temperature sensors, send it to ThingSpeak or Firebase and display it on a mobile/web dashboard. **Honesty note (README):** I did not have a NodeMCU, INA219 or DS18B20, so the hardware side was not run on a physical board. What I deliver is (a) ESP8266 firmware compile-checked with arduino-cli against the official core, (b) a Raspberry Pi version of the node, (c) a software simulator that pushes the identical payload the firmware would push, (d) a local ThingSpeak-compatible mock cloud, and (e) the responsive dashboard tested end-to-end in a headless browser. Every number in results/ is simulated.

2. Architecture

Sensing node: INA219 on I2C (D1/D2) for pack V and I, DS18B20 on OneWire (D4), 12 V aux battery through a 30k/7.5k divider on A0; 1 s sampling, 15 s averaging and upload (ThingSpeak free-tier minimum) as fields 1-8 = pack V, current, temperature, SOC, power, aux V, health-flag bitmask, state; Firebase path uses PUT latest.json + PATCH history.json. Health flags: OVER_TEMP > 45 C, UNDER_TEMP < 0 C, LOW_SOC < 20 %, OVER_CURRENT > 30 A, OVER_VOLTAGE > 4.25 V/cell, UNDER_VOLTAGE < 3.0 V/cell, AUX_LOW < 11.8 V, SENSOR_FAULT. SOC = IR-compensated OCV lookup ($V + I \cdot R_{\text{pack}}$) / cells on a 13-point table shared by firmware and simulator.

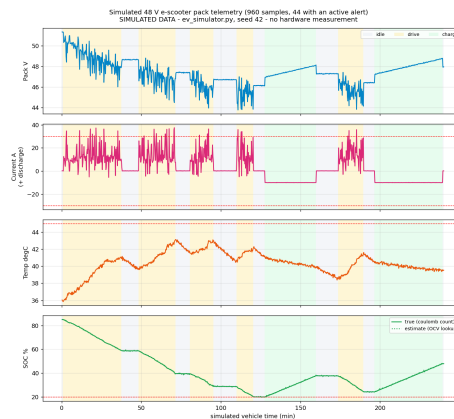


Fig. 1 - Four simulated hours of a 13s 30 Ah pack (seed 42): voltage, current, temperature, true vs estimated SOC and flags, with idle / drive / charge states shaded.

3. Method and implementation

- Firmware (nodemcu_ev_monitor.ino): Wi-Fi STA auto-reconnect, sensor-fault flag on failed reads, HTTPClient over WiFiClientSecure (setInsecure, 512 B buffers) for ThingSpeak or REST for Firebase, selected with one #define; all secrets in a git-ignored config.h with compile-safe fallbacks.
- Simulator (ev_simulator.py): seeded drive / charge / idle model of a 48 V 13s 30 Ah e-scooter pack posting to any ThingSpeak-compatible /update URL; --speed compresses vehicle time. Mock cloud (mock_cloud.py, Flask): /update, /channels/feeds.json, feeds/last.json, fields/.json with ThingSpeak's string fields and ISO timestamps, key checks (401 / 400), ring buffer, dashboard served on /.
- Dashboard (dashboard/index.html): cards, SOC ring gauge, three rolling charts, alert banner and history table polling feeds.json every 5 s; Chart.js from CDN with a built-in canvas fallback for offline use; switching to the real cloud is a URL query string.

4. Results (results/demo_summary.json - all simulated, run_demo.py seed 42)

Metric	Value	Comment
Uploads / read back via feeds.json	960 / 960	4 vehicle-hours at 15 s spacing
Pack voltage	43.76 - 51.36 V	13s pack, 3.37 - 3.95 V/cell
Current	-10.12 A (charge) ... +37.46 A	acceleration bursts trip OVER_CURRENT > 30 A
Pack temperature	35.86 - 43.13 degC	35 degC ambient, stays under the 45 degC flag
SOC estimate vs coulomb-counted truth	0.23 % mean abs error	optimistic: simulator and estimator share the pack model
Samples with an active alert	44 of 960	28 OVER_CURRENT, 16 LOW_SOC
Dashboard, headless Chromium 1366 px and 390 px	status LIVE - 0 s ago	chart engine: built-in fallback (CDN blocked in sandbox)
Firmware compile, esp8266 core 3.1.2 (recorded 2 Sept 2026)	ThingSpeak 370,332 B flash / 29,992 B RAM; Firebase 370,444 B / 30,132 B	0 warnings with --warnings all; not flashed

BOM item (hardware/bom.csv, Sept-2026 estimates +/-30 %)	Qty	Approx. INR
NodeMCU 1.0 ESP8266	1	350
INA219 breakout (0.1 ohm shunt)	1	180

KEY TAKEAWAY The full sensing-node -> cloud -> dashboard loop runs end-to-end offline against a ThingSpeak-compatible mock, with firmware and simulator sharing one tested OCV table; no board was flashed and every value is simulated.

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BOM item (hardware/bom.csv, Sept-2026 estimates +/-30 %)	Qty	Approx. INR
DS18B20 waterproof probe + 4.7 k pull-up	1	122
30 k + 7.5 k 1 % divider resistors	1 each	4
6s 18650 bench pack with BMS	1	1,500
12 V aux battery (optional)	1	600
Breadboard, jumpers, USB cable, 5 V supply	1 set	250
Total, NodeMCU path		~3,000
Raspberry Pi 4 + ADS1115 (alternative node only)	1	~4,500

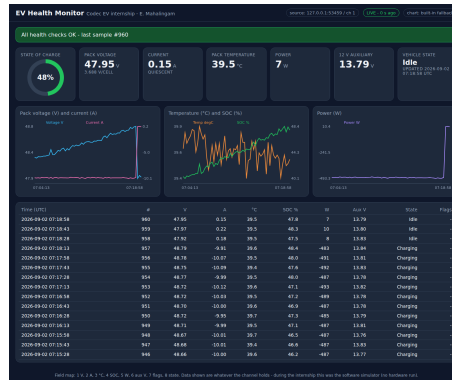


Fig. 2 - Dashboard screenshot (1366 px) against the mock cloud: LIVE status, cards, SOC gauge, rolling charts and history table.

5. Verification

Re-run here with the fixture-aware runner (pytest not installed): **17 passed, 0 failed** in 0.4 s. The mock-cloud tests run through the Flask test client, so the HTTP round trips were exercised in this session. Firmware compilation was **not** repeated here - arduino-cli is unavailable - the flash/RAM figures are the project's recorded run.

Test	Checks	Result
test_soc_lookup_monotonic_and_bounded, test_soc_inverse_round_trip, test_ir_compensation_direction	OCV->SOC monotone on a 5 mV sweep, inverse lookup round-trips, IR compensation restores rest SOC under load	3 PASS
test_ocv_table_matches_firmware, test_thresholds_match_config_example	OCV table and flag bits parsed out of the .ino equal pack_model.py; config.h.example thresholds equal Thresholds()	2 PASS
test_alert_flags, test_telemetry_field_mapping	every flag individually and multi-flag; field1..8 mapping and power = V x I	2 PASS
test_simulator_values_in_physical_ranges, test_simulator_soc_estimate_tracks_truth, test_simulator_deterministic_with_seed	6 h run inside physical ranges, visits all three states, estimate within 1 % of truth, seed-deterministic	3 PASS
test_mock_cloud_round_trip, test_mock_cloud_rejects_bad_key_and_channel, test_mock_cloud_read_key_enforced, test_mock_cloud_ring_buffer, test_simulator_posts_to_mock_cloud	post -> feeds / last / fields round trip, 401 / 400 handling, read-key enforcement, ring buffer, simulator end-to-end through Flask client	5 PASS
test_dashboard_polls_feeds_and_has_fallback, test_sample_feed_file_declared_simulated	poll URL, alert names, viewport meta, fallback chart class present; sample feed channel description says SIMULATED	2 PASS

6. Data integrity and declared substitutions

- **No hardware was tested.** Firmware is compile-verified only (recorded arduino-cli run, not repeated here). Untested: INA219 sign convention, DS18B20 timing on D4 (shared with the on-board LED), ADC calibration, TLS handshake memory on the ESP8266.
- INA219 limits (26 V bus, 3.2 A shunt) mean the default firmware measures a 6s bench pack, not a 48 V scooter pack; hardware/wiring.md gives the INA226 / external-shunt / isolated route. The simulator models the 13s pack.
- SOC from OCV alone is not a BMS-grade estimator (project 01's EKF is); setInsecure() skips certificate validation and should be replaced by fingerprint pinning; BOM prices are Sept-2026 estimates +/-30 %.

7. Learnings

Putting the OCV table, flag bits and thresholds in one Python module and unit-testing them against the .ino text is what keeps firmware, simulator and dashboard from drifting apart - the test suite fails if any of the three changes alone. Copying ThingSpeak's exact JSON shape in the mock means switching to the real cloud is a URL change, not a code change. With a board the first steps are a DMM comparison of INA219 readings on a 6s bench pack, a real ThingSpeak channel over 4G, and then the INA226 + 50 A shunt route for a road-going pack.

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